CARBON MICROPHONE M-15( )/UR

Section 2

Under Federal add "QQ-Z-325 Zinc Plating (Electrodeposited)" Under Air Force-Navy Aeronautical delete "AN-P-32 Plating Zinc." Under U. S. Army delete "71-886 Cordage CO-122-A and CO-122-B." Add "21.4 Standards, Military MIL-STD-129 Marking of Shipments."

Section 3

Paragraphs 3.2.11; 3.2.11.1; 3.2.11.2; 3.2.11.2.1; 3.2.11.2.2; 3.2.11.2.3.- Delete these paragraphs entirely and substitute the following paragraphs:

"3.2.11 Cordage.- Cordage designated as Cordage CO-122-A or B on the applicable drawing shall be in accordance with the following:

3.2.11.1 Conductors.- Shall be A.W.G. No. 22 (Cordage CO-122-A) or A.W.G. No. 20 (Cordage CO-122-B). Each conductor shall consist of a suitable number of soft copper wire strands, not larger in diameter than A.W.G. No. 34 (.0063") nor smaller in diameter than A.W.G. No. 36 (.005"). The copper strands shall be individually "tinned" and the "Tinning" material and process shall be of such a quality and nature as to protect the copper against corrosion and to provide an entirely suitable surface for soldering. The lay of the strands shall be approximately 1 inch. Over the assembled strands shall be applied a close wind of fine cotton or cellulose acetate moisture-proof treated yarn. Alternatively, the yarn may be omitted provided the conductors in the finished cordage are "free-stripping" and free from any detrimental effects which would interfere with normal soldering of the strands of the conductors.

3.2.11.2 Central Cores, Fillers, Binders, Etc.- All cotton or other textiles used as a central core, filler, binder, etc., shall be mildew-proof treated in accordance with Specification MIL-M-5658 except where natural rubber insulation is used in which case some other approved treatment shall be used. The impregnating material shall not affect the electrical or physical characteristics of the cordage.

3.2.11.3 Insulation.- The insulation for each conductor shall consist of a suitable compound of rubber or rubber substitute, homogeneous in character, tough, elastic and applied concentrically about the conductor, fitting closely thereto. The minimum wall thickness at any point along the insulation shall be not less than 0.010 inch; the desirable thickness is 0.020 inch.

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3.2.11.4 Coloring of Insulation.- The insulation on the individual conductors is to be colored red, white, and black respectively. The colors may be obtained by the use of compound colored throughout or by the use of a thin layer of colored compound applied over the natural colored compound. If the latter method is used the colored compound shall form an integral part of the insulation.

3.2.11.5 Cabling.- Three insulated conductors (one red, one white and one black) shall be spiralled around a central core with a lay of approximately 0.4 inch. The central core shall have a minimum diameter of 1/16" and a breaking strength of not less than 30 pounds and shall be of such a design, quality and nature as to be entirely suitable for this application. The central core shall not abrade or otherwise damage the insulation on the conductor when the cordage is bent or stretched. The cabling shall be such as to provide relatively free movement of the cabled conductors over the central core, when the finished cordage is bent or stretched. The conductors shall not be tightly wrapped on the central core in the finished cordage. Filling or serving may be used provided that flexibility consistent with cordage of this type is maintained.

3.2.11.6 Jacket.- After cabling, the assembly shall be covered over all with a jacket of black compound of rubber or rubber substitute so compounded and processed as to provide long life under hard service conditions. The jacket shall have a minimum thickness of 0.025 inch and shall not adhere or stick to the conductor insulation. Optionally, the contractor may apply a thin cotton wrapping over the assembled conductors before the jacket is applied, provided that suitable flexibility is maintained in the finished cordage. After vulcanization, the jacket shall be a firmly-bound, tough, elastic, and homogeneous mass. The outer jacket of the finished cordage shall have a reasonably smooth surface, that is a surface at least as smooth as would result from a continuous vulcanization process. The jacket shall show no evidence of "blooming" as a result of being subjected to a temperature of 70°C. (158°F) for a continuous period of two hours.

3.2.11.7 Outside Diameter.- The outside diameter at any point along the finished cordage shall be not greater than 0.290 inch nor smaller than 0.270 inch; the desirable diameter is 0.280 inch.

3.2.11.8 Insulation and Jacket Compounds.- The insulation and the jacket of the finished cordage shall meet the requirements of Specification No. 71-4945 except as specified herein.

3.2.11.9 Temperature Requirements:

3.2.11.9.1 Operating Range.- The cordage shall be flexible and resilient throughout the temperature range of minus 55 degrees C to plus 85 degrees C.

3.2.11.9.2 Cold Bend.- The cordage shall be capable of meeting the cold bend requirements of Specification No. 71-4945 with the cold chamber temperature maintained at minus 60 degrees C, plus or minus 2 degrees C.

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3.2.11.9.3 Flexing.- The cordage shall be capable of withstanding at least 7000 flexing cycles without showing any evidence of:

(1) Damage to the jacket of the cordage.
(2) Damage to the insulation of the individual conductors.
(3) Electrical discontinuity.

Paragraph 3.17.- Under "d" delete AN-P-32 and replace by QQ-Z-325.

Section 4

Paragraph 4.2.1 Preproduction Test Sample.- In the first sentence after "microphones" add "and 30 feet of cordage."

Paragraph 4.2.1.1.- Delete and replace by the following: "4.2.1.1 The preproduction samples shall be in all respects truly representative of the items the contractor proposes and expects to furnish in quantity production."

Paragraph 4.2.1.2 Test Data With Samples.- Delete and replace by the following: "4.2.1.2 Test Data With Samples.- Laboratory test data for the microphone and cordage represented by the samples submitted shall be furnished to the designated Laboratory at the time the samples are submitted. The test data submitted shall show the results of complete tests performed on microphones and a sample lot of cordage represented by the sample submitted. Test data is not required for items that have been approved previously by qualification approval, such as the plug. If the item or part has been approved previously, the date of approval, the name of the approving agency, and the previous contract or order number involved shall be furnished.

4.2.1.3 Informational Data.- The sample length of cordage and the microphones shall be accompanied by adequate informational data, including the following:

(1) A list of the materials used in assembly of the microphone and cordage.

(2) Details regarding the cordage assembly.

(3) The Government order number involved ("File No." in contract), if any.

(4) Prime contractor's name and address.

(5) Manufacturer's name and address.

(6) Name and address of the Laboratory which conducted the tests referred to in the above paragraph on Test Data With Samples.

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Add the following new paragraphs:

4.2.2.15 Electrical Tests of Cordage.- The finished cordage, in convenient lengths, shall be immersed in water for a period of not less than twelve hours after which the following tests shall be made while the cordage is still immersed and the water grounded. During these tests neither the extreme ends of the cordage jacket nor the bared conductors shall be immersed in the water.

4.2.2.15.1 Dielectric Strength.- The cordage shall withstand, without breakdown, the application of a 60-cycle alternating potential of 500 volts, effective, between each conductor and the outer conductors grounded, for one (1) minute. The energy source for this test shall have a capacity of not less than 5 kilowatts.

4.2.2.15.2 Insulation Resistance.- This test shall be made immediately after the cordage has successfully withstood the dielectric strength test. The linear insulation resistance at 60°F (15.6°C) shall be not less than 1000 megohms per thousand feet between conductors and between each conductor and ground. The measurement shall be made after one minute electrification with a continuous e.m.f. of any convenient voltage between 100 and 500 volts, the conductors being maintained negative to the water.

4.2.2.15.3 Conductor Resistance.- This test shall be accomplished by connecting the black and red conductors together at one end of each length of cordage to form a loop or close circuit. A direct current of three amperes shall then be passed through the loop formed and the resistance measured at the end of one minute. This same procedure shall be followed with the black and white conductors connected together. The resistance per conductor, per hundred feet of Cordage CO-122-A shall not exceed two and one-half ohms; the resistance per conductor, per hundred feet of cordage CO-122-B, shall not exceed two ohms.

4.5.1.5 Production Inspection of Cords:

4.5.1.5.1 Mechanical Inspection.- Each complete cord shall be given a thorough mechanical inspection; special attention being given to neatness and thoroughness of soldering, twisting, and tinning of the conductor strands and to the condition and spiral lay of the conductors at the ends of the outer jacket.

4.5.1.5.2 Electrical Tests.- Following the above test, each cord shall withstand 250 volts direct current or peak alternating current applied between conductors for two seconds. Each cord shall be tested for continuity.

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Section 5

Paragraph 5.1 General.- Delete and replace by the following:

5.1 General.- Packaging and packing shall be in accordance with Method II of Specification NO. 7200.

5.1.1 Marking of Shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The following information shall be included directly after the nomenclature: Specification MIL-C-6629(USAF).

THIS AMENDMENT FORMS A PART OF AND SHOULD BE FILED WITH SPECIFICATION MIL-C-6629(USAF) DATED 1 NOVEMBER 1951.

(Copies of this amendment and the basic specification may be obtained from the Commanding General, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio.)

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